

Dock Book II

Forty-five fresh Domain 3 questions — none seen in the Loading Dock or its eight explained parts. Same distribution, same shapes, same families. Sit it cold; the key is sealed until Q45.

DISTRIBUTION	3.1 × 5 · 3.2 × 6 · 3.3 × 5 · 3.4 × 6 · 3.5 × 7 · 3.6 × 6 · 3.7 × 5 · 3.8 × 5 — new territory carries the weight
SHAPES	14 head-to-head pairs · 10 number-decided stems · 4 D1 crossovers
RULES	underline the deciding clause first · 120 seconds each · ~90 minutes · key sealed · every miss gets a ledger row
ANSWER LETTERS	deliberately mixed — the first book's all-B pattern does not repeat here
ON TRACK	34+/45 with no objective below 60%; a VG-heavy miss profile → one more glossary pass, not a re-read

DOCK BOOK II · INSTRUCTIONS, THEN FORTY-FIVE QUESTIONS

Before you start

Same rules as the first book. **Underline the deciding clause before you read the options.** One hundred and twenty seconds per question, one sitting, about ninety minutes. Mark an answer for every question — an unmarked question is a miss with no ledger row. The key is sealed until Q45.

What is different from the first book. Every scenario is new: no question re-skins a Dock Book stem. Correct letters are mixed across A-D. The distractor families are the same eight — **OE** over-engineered · **VG** vocabulary gap · **TR** trade-off reversed · **CM** constraint missed · **HL** human loop skipped · **SF** silent failure · **SD** sounds-diligent · **OP** over-permissive — and every miss in the key names its family.

Objective	Count	Shapes you will meet
3.1 Capability bloat	5	two number-decided, one head-to-head
3.2 AuthN / AuthZ	6	one head-to-head, one confused-deputy stem
3.3 Trade-offs, justified	5	two number-decided, one head-to-head
3.4 Observability	6	one number-decided, one D1 crossover (cost)
3.5 Chunking & indexing	7	one head-to-head, two number-decided
3.6 Retrieval by shape	6	one head-to-head, one D1 crossover (decomposition)
3.7 Integration mechanism	5	one head-to-head
3.8 Progressive vs monolith	5	two head-to-head, one number-decided, one D1 crossover

© THE HABIT THAT SCORES

Read every stem twice: once for the constraint it FIXES (an SLA, a floor, a ceiling, a data shape, a client count), once for the number it hands you. Ten stems are decided by arithmetic you can do in the margin. When two options both satisfy the stated constraint, the tie-break is the currency the stem did NOT free up.

Objective 3.1

Q1 · objective 3.1 · number-decided

Six weeks after launch, the logs show `convert_currency` called 0 times and `get_customs_duty` (added on ticket evidence) called ~300 times a week. A PM proposes keeping both 'since a schema is only ~260 tokens'. At 40,000 calls/day, the cost of keeping `convert_currency`, and the verdict:

- A. ~10.4M tokens/day; trim `convert_currency`, keep `get_customs_duty`
- B. Trim both to a single 'finance' tool that wraps currency and duty
- C. Keep both; trimming after launch confuses users who have seen the capability
- D. ~1M tokens/day; acceptable insurance for a currency feature customers may want

Answer: ____ · the tell I underlined: _____

Q2 · objective 3.1 · head-to-head

Head-to-head. Two ETA tools, `get_eta` and `get_carrier_eta`, overlap; the model picks the wrong one 11% of the time. (B) merge them into one tool with a carrier parameter that is optional; (C) keep both and add to the system prompt 'use `get_carrier_eta` when the carrier is known'. Pick.

- A. B — merge into one tool with a carrier parameter
- B. C — the instruction encodes the rule and keeps the tools independent
- C. Either; 11% is within tolerance for ETA questions
- D. C, plus a third tool that decides which of the two to call

Answer: ____ · the tell I underlined: _____

Q3 · objective 3.1

The refund-desk route carries 5 tools (including `process_refund`); the dispatcher-draft route carries 4; the router carries 2. A platform lead proposes one unified 9-tool belt on every route 'for consistency'. The 3.1 verdict:

- A. Reject: per-route belts are the bloat control and the security boundary
- B. Accept: one belt is simpler to maintain and schemas are cached anyway
- C. Accept, but add a system-prompt rule that draft sessions must not refund
- D. Reject, and cut every route to 2 tools to be safe

Answer: ____ · the tell I underlined: _____

Q4 · objective 3.1

A tool's docstring reads 'Handles orders.' The model calls it for order creation, order status and order cancellation, and it only does status. Best fix:

- A. Rewrite the docstring: what it does, and when NOT to use it
- B. Rename the function to `handle_orders_v2`
- C. Fine-tune the model on the correct call pattern
- D. Add three examples of correct calls to the system prompt

Answer: ____ · the tell I underlined: _____

Q5 · objective 3.1 · number-decided

count_tokens reports 3,420 input tokens with the 12-tool belt and 1,740 with the 5 tools the logs say carry the traffic. Daily volume is 25,000 calls. The daily belt tax of the 7 idle tools, and the honest note:

- A.** ~4.2M tokens/day; small enough to ignore
- B.** ~85M tokens/day, because schemas are counted twice per turn
- C.** Cannot be known without running the 25,000 calls
- D.** ~42M tokens/day — and the probe that priced it was free

Answer: ____ · the tell I underlined: _____

Objective 3.2**Q6 · objective 3.2**

A new customs-broker integration receives an API key, authenticates every time, and the audit log shows it reading refund histories it never needs. There is no scope table anywhere — permissions live in three people's heads. The first move:

- A.** Rotate the broker's API key immediately
- B.** Add 'only read customs data' to the broker integration's prompt
- C.** Write the per-caller scope table down, then narrow the broker to customs:read
- D.** Encrypt the refund history table at rest

Answer: ____ · the tell I underlined: _____

Q7 · objective 3.2

Dispatchers sign in to the UI by SSO. The UI calls the KAVI agent, which calls every MCP tool as the single 'kavi-agent' service account. The refund desk wants per-dispatcher refund limits. What is missing?

- A.** A larger model that reasons about who is asking
- B.** Nothing — the UI already knows who the dispatcher is and can enforce the limit
- C.** Propagate the dispatcher's identity to the tool and authorize on it
- D.** A per-dispatcher system prompt stating their limit

Answer: ____ · the tell I underlined: _____

Q8 · objective 3.2

A workbook exercise says the SOP server running over stdio on Arjun's laptop 'must be secured with OAuth before use'. The accurate correction:

- A.** No OAuth needed: a stdio server inherits the OS user's identity
- B.** Correct as written; every MCP server needs OAuth
- C.** OAuth is deprecated in the 2026-07-28 spec
- D.** Correct, but an API key is enough for stdio

Answer: ____ · the tell I underlined: _____

Q9 · objective 3.2 · head-to-head

Head-to-head, both found in the same audit. (B) a token rotated every 90 days carrying scope '*'; (C) a static token, unrotated for three years, carrying only status:read. Which is the worse gap, and why?

- A.** B — a wildcard is a live key to every bay; fix both, B first
- B.** Equal; rotate C and rotate B faster
- C.** Neither is a gap while no incident has occurred
- D.** C — three years unrotated is the textbook finding

Answer: ____ · the tell I underlined: _____

Q10 · objective 3.2

Since Monday, the denials panel shows ~400 denials/day: caller kavi-agent, tool process_refund, all from the drafting route. No refunds went out. The reading and the fix:

- A.** Silence the panel; the denials are noise since no refund went out
- B.** An external probe — rotate kavi-agent's key
- C.** A misconfiguration: remove process_refund from the drafting route's belt
- D.** Grant refund:write to kavi-agent so the drafts stop failing

Answer: ____ · the tell I underlined: _____

Q11 · objective 3.2

The MCP resource sop://214 contains internal escalation phone numbers. A customer-facing chatbot identity holds sop:read so it can quote procedures. The 3.2 gap:

- A.** Move the phone numbers into a tool instead of a resource
- B.** Instruct the chatbot never to reveal phone numbers
- C.** Resource scopes are too coarse: split the content or the scope
- D.** None — resources are read-only, so they need no authorization

Answer: ____ · the tell I underlined: _____

Objective 3.3**Q12 · objective 3.3 · number-decided**

Agent-tier investigations cost ₹38/case on an Opus-class model and are 2% of volume; delay drafts cost ₹1.20 and are 98%. The blended ceiling is ₹6 per exception. A cost review proposes downgrading the agent tier 'to fit the ceiling'. The number, and the verdict:

- A.** Raise the ceiling to ₹40 to cover the agent tier
- B.** Blended ≈ ₹1.94 — already under ₹6; the downgrade buys nothing
- C.** Blended ≈ ₹19.60; downgrade needed
- D.** Downgrade anyway; cheaper is always safer under a ceiling

Answer: ____ · the tell I underlined: _____

Q13 · objective 3.3

The SOP-answer path has a 4 s SLA; p95 is 3.85 s. A proposal raises top_k from 20 to 40 'for better accuracy'; measured: +350 ms, and answer quality did not improve beyond k=20. Verdict:

- A.** Accept and raise the SLA to 4.5 s
- B.** Reject, and lower top_k to 5 to buy back latency
- C.** Accept — more context is always safer for SOP answers
- D.** Reject — breaches the SLA at p95, and no measured gain past k=20

Answer: ____ · the tell I underlined: _____

Q14 · objective 3.3 · number-decided

Three carrier ETA calls run one after another: 0.9 s, 1.1 s and 0.8 s. The dispatcher SLA is 2 s. The 3.3-correct fix:

- A.** Drop the slowest carrier from the fan-out
- B.** Run the three calls concurrently — wall clock $\approx$ 1.1 s
- C.** Move the ETA calls to a faster model
- D.** Cache all ETAs for an hour

Answer: ____ · the tell I underlined: _____

Q15 · objective 3.3 · head-to-head

Head-to-head on the refund path, which has an accuracy floor of $\geq 99\%$ on threshold detection and no latency SLA tighter than 5 s. (B) small model: 97.5% correct, 400 ms. (C) large model: 99.3% correct, 1,300 ms. Pick.

- A.** C — the floor is binding and B cannot clear it; 1.3 s is slack
- B.** B, with a retry when confidence is low
- C.** B — three times faster, and 1.5 points is negligible
- D.** C, only if latency is removed from the SLA

Answer: ____ · the tell I underlined: _____

Q16 · objective 3.3

Which justification for a dial change is exam-grade?

- A.** 'Refunds err at $\sim$ ₹1,800 and are 9% of volume $\rightarrow$ large model there; blended stays $\leq$ ₹6'
- B.** 'The team is more comfortable with the larger model'
- C.** 'A more capable model gives better accuracy, which matters for refunds'
- D.** 'Industry best practice is to use the largest model for financial decisions'

Answer: ____ · the tell I underlined: _____

Objective 3.4**Q17 · objective 3.4 · number-decided**

The retrieve stage shows p50 = 200 ms and p99 = 6.1 s. The SLA is 'p99 under 3 s'. The dashboard shows p50 only, and it is green. The 3.4 reading:

- A.** Rewrite the SLA in p50 terms so the dashboard matches
- B.** Add the p99 panel — the SLA is written in p99 and is breached at the tail
- C.** Healthy — p50 is well inside 3 s
- D.** Alert on every request over 3 s

Answer: ____ · the tell I underlined: _____

Q18 · objective 3.4

Log lines read like 'retrieved chunks for ticket in 310ms' and 'draft done, 880 ms'. The team wants a p95-by-stage dashboard. What must happen first?

- A.** Sample 10% of lines and read them by hand weekly
- B.** Buy a dashboard product that parses free text with regex
- C.** Add a dashboard now; refine the logs later
- D.** Convert to structured JSON with stage, ms and trace fields first

Answer: ____ · the tell I underlined: _____

Q19 · objective 3.4

Trace ids are stamped on every stage's log line, but the carrier ETA call runs in a background worker that starts with a fresh context. The slow span everyone suspects never appears in any trace. The gap:

- A.** Propagate the trace id into the background call so its span joins the trace
- B.** Increase log retention so the span can be found later
- C.** Move the carrier call inline to avoid background workers
- D.** The carrier API is fast; the suspicion is wrong

Answer: ____ · the tell I underlined: _____

Q20 · objective 3.4

Which metric label set is safe at 40,000 tickets a day?

- A.** {dispatcher_email, stage} — to see who is slow
- B.** {trace_id} — so every metric point links to its trace
- C.** {stage, consignment_id} — for per-ticket drill-down
- D.** {stage, carrier, route}

Answer: ____ · the tell I underlined: _____

Q21 · objective 3.4

Which alert rule is 3.4-correct for the dispatcher-facing path?

- A.** 'Any log line containing the word error'
- B.** 'Cache hit rate below 92% for one minute'
- C.** 'p95 total latency > 3.5 s sustained for 5 minutes'
- D.** 'Any single request > 3 s'

Answer: ____ · the tell I underlined: _____

Q22 · objective 3.4 · D1 crossover

A D1 crossover. The monthly bill rose 40%. Every stage's log line carries tokens and cost_pc (paise), stamped with route and stage. How do you find the cause in minutes?

- A.** Add a cost dashboard first, then instrument later
- B.** Cut every route to the smallest model until the bill drops
- C.** Ask the model vendor for an itemised invoice
- D.** Aggregate cost_pc by route and stage from the register

Answer: ____ · the tell I underlined: _____

Objective 3.5**Q23 · objective 3.5**

The corpus is 2,000 scanned carrier contracts: 6M tokens, no headings, dense legal prose, queried for specific clauses. The 3.5 chunking:

- A.** Sentence-level chunks for clause precision
- B.** One chunk per contract; clauses need their whole context
- C.** Fixed-size with overlap and a situating sentence per chunk
- D.** Heading-aware chunks; contracts have sections

Answer: ____ · the tell I underlined: _____

Q24 · objective 3.5

Hindi-English SOPs: a query for 'e-way bill' misses chunks written as 'ए-वेय बिल'. The 3.5 fix:

- A.** Drop keyword search; embeddings handle languages
- B.** Translate every incoming query into Hindi before searching
- C.** Raise top_k so the Hindi chunks eventually appear
- D.** Situate every chunk in English at ingest; let the semantic channel bridge the script

Answer: ____ · the tell I underlined: _____

Q25 · objective 3.5 · head-to-head

A 60k-token rate-and-template pack, stable for quarters, needed on nearly every call. Head-to-head: (B) prompt caching of the whole pack; (C) chunk, index and retrieve top-20 per call (~3.6k tokens + ~180 ms). Pick.

- A.** C — 3.6k beats 6k on the token column
- B.** B — cache the whole pack; ~6k token-equivalents, no retrieval hop
- C.** B, but re-chunk anyway for when the pack grows
- D.** C, because caching is unreliable across model tiers

Answer: ____ · the tell I underlined: _____

Q26 · objective 3.5

After re-chunking on headings, the miss ledger shows a new pattern: the right section is retrieved at rank 7, but only the top 5 chunks reach the model. The fix:

- A.** Re-chunk again with smaller sections
- B.** Swap the embedding model
- C.** Pass top-20 and add reranking — the fault is depth, not shelving
- D.** Add a system-prompt note about which section matters

Answer: ____ · the tell I underlined: _____

Q27 · objective 3.5 · number-decided

Overlap is set to 400 tokens on 500-token chunks. The index has doubled and the top-20 shortlist is mostly near-duplicates. The 3.5 reading:

- A.** Raise top_k to 40 to get past the duplicates
- B.** Overlap is a dial turned too far; set it to a few dozen tokens and re-index
- C.** Remove overlap entirely
- D.** Overlap is free insurance; keep it

Answer: ____ · the tell I underlined: _____

Q28 · objective 3.5

The index is rebuilt weekly, but chunks carry no version. A dispatcher is read a procedure repealed two weeks ago, with a confident citation. Required fix:

- A.** Nothing — weekly is fresh enough
- B.** Write a version label into every chunk and drop superseded chunks on re-index
- C.** Re-embed on every edit in real time
- D.** Tell the model to prefer newer-sounding text

Answer: ____ · the tell I underlined: _____

Q29 · objective 3.5 · number-decided

400 SOPs → ~3,600 chunks. A PM rejects contextual retrieval: 'writing a situating sentence per chunk with a model runs on every query — 40,000 calls a day'. The correction:

- A.** It runs per query but is worth the 35–49% failure cut
- B.** Situating runs at ingest (~3,600 calls per weekly rebuild), not per query
- C.** Correct, unless the situating model is the smallest tier
- D.** Correct — contextual retrieval doubles query cost

Answer: ____ · the tell I underlined: _____

Objective 3.6**Q30 · objective 3.6**

'What is the HSN code for lithium batteries?' — the answer is one row in a 12,000-row tariff table. The 3.6 machine:

- A.** A structured lookup (or exact keyword match) against the table
- B.** Ask the model; it knows tariff codes
- C.** Semantic search over an embedded copy of the table
- D.** Hybrid RAG with reranking

Answer: ____ · the tell I underlined: _____

Q31 · objective 3.6

'Show me the trend of customs holds per carrier this month.' Routed to hybrid RAG, it returns a paragraph with numbers and SOP citations. What happened, and the route?

- A.** Fine — the citations prove the numbers
- B.** Raise top_k so all the hold records are retrieved
- C.** Index the TMS rows so RAG can see them
- D.** RAG composed the numbers; route trend/per-month language to SQL over TMS rows

Answer: ____ · the tell I underlined: _____

Q32 · objective 3.6 · D1 crossover

'Why do BlueDart parcels to Guwahati keep getting stuck, and what should I do?' The question mixes shapes. The 3.6 handling:

- A.** Decompose: an aggregate for 'keep getting stuck', hybrid RAG for 'what should I do'
- B.** Structured lookup on BlueDart's Guwahati lane
- C.** Hybrid RAG for the whole question; it is mostly prose
- D.** Escalate to an agent to reason it out

Answer: ____ · the tell I underlined: _____

Q33 · objective 3.6 · head-to-head

Head-to-head for dispute emails. Query: 'What did BlueDart promise on 14 Aug about KF-2481?' (B) semantic-leaning hybrid over raw email text; (C) keyword-leaning hybrid over chunks situated with carrier, consignment and date metadata, filtered on that metadata. Pick.

- A.** C — keyword-lean with a metadata filter; an id and a date are exact strings
- B.** Neither — summarise each email and search summaries
- C.** B — dispute language is fuzzy; meaning beats words
- D.** Either; both channels see the same emails

Answer: ____ · the tell I underlined: _____

Q34 · objective 3.6

A reranker was added to the exact-code path ('HSN 8471', section numbers). Measured: +200 ms, no change in answer quality; on the procedure-prose path it cut failures 67%. The 3.6 dial:

- A.** Keep it everywhere for consistency
- B.** Reranking ON for prose paths, OFF for exact-code paths
- C.** Remove it everywhere; 200 ms is too much
- D.** Replace the reranker with a larger embedding model

Answer: ____ · the tell I underlined: _____

Q35 · objective 3.6

'How many times has KF-2481 been delayed this quarter?' — an id and a count in one question. The route:

- A.** Hybrid RAG over SOPs mentioning delays
- B.** An aggregate query filtered by the id — SQL WHERE consignment = KF-2481
- C.** Ask the carrier's booking agent
- D.** Structured lookup — the id pattern says exact world

Answer: ____ · the tell I underlined: _____

Objective 3.7**Q36 · objective 3.7**

Six internal teams need the same customs-duty tool, and the duty rules (hence the tool's inputs) change monthly. The mechanism:

- A.** Each team calls the duty API directly
- B.** A shared Python package each team vendors
- C.** One MCP server; six clients discover the changed schema via tools/list
- D.** Agent-to-agent between the teams' assistants

Answer: ____ · the tell I underlined: _____

Q37 · objective 3.7

The customs_triage procedure changes every fortnight; forty dispatchers keep pasting last month's version into their chats. The MCP primitive:

- A.** A resource holding the procedure
- B.** Elicitation, so the server can ask which version they want
- C.** A tool that returns the procedure text
- D.** A prompt — the procedure lives once on the server as a versioned template

Answer: ____ · the tell I underlined: _____

Q38 · objective 3.7 · head-to-head

Head-to-head. IndiaPost offers one REST endpoint. Today the control tower AND KAVI both need it, and a third client is planned. (B) wrap it as an MCP tool now; (C) call the REST endpoint directly from each client. Pick.

- A.** C now, B when the tenth client appears
- B.** B — two clients today and a third planned pays for the connector
- C.** C — one endpoint is too small for MCP
- D.** B, and migrate every other integration to MCP the same sprint

Answer: ____ · the tell I underlined: _____

Q39 · objective 3.7

Mid-booking, the MCP server needs the dispatcher to confirm a pickup slot before it commits. Under the current spec, the server uses:

- A.** A logging notification the UI turns into a dialog
- B.** Sampling — the server asks the client's LLM to decide
- C.** A second tool call from the client
- D.** Elicitation — the server asks the user mid-task

Answer: ____ · the tell I underlined: _____

Q40 · objective 3.7

Delhivery's booking agent negotiates slots. Which brief is agent-to-agent-correct?

- A.** The consignment id only; the agent knows its own rules
- B.** A stateless call: book_slot(consignment) → slot
- C.** A full dump of the TMS row and the SOP corpus
- D.** Consignment, constraints, an authority limit, a turn budget; replies are proposals to verify

Answer: ____ · the tell I underlined: _____

Objective 3.8**Q41 · objective 3.8**

12 tools × 260 tokens on the drafting route; the logs show 10 of the 12 are used on almost every request. A 'catalogue and fetch' migration is proposed. Verdict:

- A.** Drop the 2 rare tools and call it progressive
- B.** Adopt — progressive is always leaner
- C.** Reject — keep full schemas in the cached prefix; 10 of 12 are always used
- D.** Adopt for the 2 rare tools only, catalogue the other 10

Answer: ____ · the tell I underlined: _____

Q42 · objective 3.8 · head-to-head

A 150k-token policy pack is stable for quarters but consulted by only ~30% of calls. Head-to-head: (B) monolith behind prompt caching; (C) retrieve top-20 from it when the router says policy. Pick.

- A.** B — cache reads are 90% off, so the waste is small
- B.** Neither; summarise the pack to 5k tokens
- C.** C — retrieve on demand; 70% of calls would carry 150k tokens for nothing
- D.** B — under 200k and stable, so cache it

Answer: ____ · the tell I underlined: _____

Q43 · objective 3.8

Context editing clears old tool results at 100k tokens. An investigation lost the get_shipment_status results it was reasoning over. The current-API fix:

- A.** Summarise the status results client-side each turn
- B.** Name the tool in exclude_tools; let the memory tool save essentials first
- C.** Raise the trigger to 200k
- D.** Disable context editing for the agent tier

Answer: ____ · the tell I underlined: _____

Q44 · objective 3.8 · number-decided

A Haiku-class route marks a 2,000-token preamble for caching. Cache hits: zero, ever. The reason:

- A.** The preamble must be marked on the user turn, not the system prompt
- B.** Haiku-class models do not support caching
- C.** 2,000 tokens is below Haiku 4.5's 4,096-token minimum cacheable prefix
- D.** The cache expired between calls

Answer: ____ · the tell I underlined: _____

Q45 · objective 3.8 · D1 crossover

D1 crossover, head-to-head. A nightly batch of 400 cases needs a 30k-token stable rate card AND the relevant SOP sections (corpus 1.4M). (B) rate card in a cached prefix + per-case hybrid retrieval top-20; (C) fetch the rate card progressively per case + per-case retrieval. Pick.

- A.** B, but put the whole SOP corpus in the prefix too
- B.** Neither; run the batch without the rate card
- C.** C — one mechanism for everything is simpler
- D.** B — cache the small stable card; retrieve the large case-specific corpus

Answer: ____ · the tell I underlined: _____

SEALED

THE KEY

Do not turn this page until all forty-five answers are marked.

Score by objective on the grid overleaf. 34+/45 with no objective below 60% = on track.

Score by objective

Objective	Questions	Score	Family of each miss	Below 60%?
3.1	1, 2, 3, 4, 5	/ 5		
3.2	6, 7, 8, 9, 10, 11	/ 6		
3.3	12, 13, 14, 15, 16	/ 5		
3.4	17, 18, 19, 20, 21, 22	/ 6		
3.5	23, 24, 25, 26, 27, 28, 29	/ 7		
3.6	30, 31, 32, 33, 34, 35	/ 6		
3.7	36, 37, 38, 39, 40	/ 5		
3.8	41, 42, 43, 44, 45	/ 5		
Total		/ 45		

The miss ledger — one row per miss, ten minutes

Pattern rule: three VG rows → one more glossary pass beats any re-read. Three CM rows → slow down and underline. Three TR rows → you are spending the fixed currency; re-read Part 6's tell. A wrong cell on the A3 matrix → redraw A3 from memory before the next sitting.

Date	Q#	Obj	Family	The tell I skipped (one line, my words)	Fixed?

The key — answers, tells, and why each distractor fails

Distractor families (why a wrong option is wrong): **OE** over-engineered · **VG** vocabulary gap · **TR** trade-off reversed · **CM** constraint missed · **HL** human loop skipped · **SF** silent failure · **SD** sounds-diligent · **OP** over-permissive.

Q1 A · obj 3.1

The tell: 0 times ... ~260 tokens ... 40,000 calls/day

Why A: ~10.4M tokens/day of belt tax for a tool the logs never asked for — trim it, keep get_customs_duty (the logs asked for that one weekly). $260 \times 40,000 = 10.4$ million tokens a day. Membership is decided by the logs: one tool earned its place, one did not.

B [OE] a mega-tool hides two schemas in one vaguer one; it also loses the duty tool's sharp description

C [SD] users see answers, not schemas; nobody misses a tool with zero calls

D [VG] an order of magnitude low, and 'may want' is not a log line

Q2 A · obj 3.1

The tell: overlap ... wrong one 11% of the time

Why A: B — one sharp door with a parameter; the wrong-pick rate cannot exist when there is nothing to pick between. Lookalikes are a description defect; the fix is fewer, sharper tools. A parameter absorbs the distinction; a prompt only pleads for it.

B [SD] instructions do not unload a belt; the model still faces two lookalike cards on every call

C [CM] 11% wrong tool at 40,000 calls/day is 4,400 wrong ETAs a day

D [OE] a doorman over two doors — the orchestration ghost at tool scale

Q3 A · obj 3.1

The tell: one unified 9-tool belt on every route

Why A: Reject: per-route belts are the bloat control AND the security boundary — the draft route must not even see process_refund. A tool is a token cost, a confusion cost and a door. Per-route belts keep all three small; unifying them pays every route for every route's tools and hands the draft route a refund door.

B [CM] caching discounts the token bill; it does nothing for wrong-tool rates or for the door

C [SD] persuasion guarding a door the config just opened (Part 3's whole lesson)

D [TR] over-learning the trim: the refund desk needs its five

Q4 A · obj 3.1

The tell: docstring reads 'Handles orders.'

Why A: Rewrite the docstring to say exactly what it does and when NOT to use it — the docstring IS the interface the model reads. In FastMCP the description comes from the docstring; the model decides from that sentence. A vague card is a mis-dialled call.

B [VG] the name is not the defect; the description is

C [OE] training-scale machinery for a one-sentence defect

D [SD] patching the prompt for a defect that lives in the tool card

Q5 D · obj 3.1

The tell: 3,420 ... 1,740 ... 25,000

Why D: ~42M tokens/day — $(3,420 - 1,740) \times 25,000$ — and the probe that produced these numbers was free. $1,680 \times 25,000 = 42,000,000$. count_tokens priced both belts without a paid call; the arithmetic ends the meeting.

A [VG] slipped an order of magnitude

B [VG] invented mechanism; the count is per request

C [CM] count_tokens exists precisely so you never have to

Q6 C · obj 3.2

The tell: authenticates every time ... reading refund histories it never needs ... no scope table

Why C: Write the per-caller scope table down, then narrow the broker to customs:read — what cannot be tabled cannot be audited. A working login with an unexplained data reach is an authZ gap. The map nobody drew is the meta-lesson: document, then narrow.

A [CM] a fresh key to every bay is still a key to every bay

B [SD] persuasion where a scope check belongs

D [SD] clever cryptography for a permission problem; the broker still decrypts it legitimately

Q7 C · obj 3.2**The tell:** calls every MCP tool as the single 'kavi-agent' service account**Why C:** The end-user's identity is lost at the agent hop — propagate the dispatcher's identity to the tool and authorize on it (the confused-deputy gap). Authorization can only be scoped to identities that reach the check. If every call arrives as kavi-agent, per-dispatcher limits are impossible by construction.**A [VG]** judgement is not a control**B [SD]** a UI-side check is a client-side promise; the tool must enforce**D [SD]** the persuasion gate, with extra steps**Q8 A · obj 3.2****The tell:** over stdio on Ariun's laptop**Why A:** A stdio server inherits the identity of the OS user who launched it — OAuth belongs on the remote Streamable HTTP flavour. AuthN is transport-shaped: stdio = OS-level identity, one local client; Streamable HTTP = standard HTTP auth, OAuth recommended.**B [VG]** transport decides authN; stdio has no network to authenticate over**C [VG]** sampling and logging were deprecated; OAuth is recommended**D [VG]** there is no bearer to present on a local pipe**Q9 A · obj 3.2****The tell:** scope '*' ... only status:read**Why A:** B — a wildcard is a live key to every bay today; C is stale hygiene but can only read statuses. Fix both, B first. Rotation is about how long a stolen key works; scope is about what it works ON. The Refund Heist's ghost is the wildcard, not the calendar.**B [CM]** rotating a wildcard faster still issues a wildcard**C [SF]** the audit IS the incident**D [TR]** prioritises the calendar over the bays**Q10 C · obj 3.2****The tell:** caller kavi-agent ... all from the drafting route ... no refunds went out**Why C:** The gate is working; something on the drafting route now offers process_refund to the model — remove the tool from that route's belt (3.1 + 3.2). Denials from your own agent are a misconfiguration signal, not an attack: the model keeps reaching for a door the belt should not show it. Trim the belt; leave the scope denied.**A [SF]** the panel just caught a config regression on day one**B [CM]** the caller is your own agent on your own route; read the identity on the line**D [OP]** opening the door because someone keeps knocking**Q11 C · obj 3.2****The tell:** customer-facing ... holds sop:read ... internal escalation phone numbers**Why C:** Resource scopes are too coarse — split internal SOP content (or the scope) so the customer-facing identity cannot read escalation details. Least privilege applies to resources as much as tools: 'read-only' is not 'harmless'. What a caller can read into context, it can leak into an answer.**A [VG]** changes the primitive, not who may reach it**B [SD]** persuasion guarding content the identity is allowed to fetch**D [OP]** read-only data can still be exfiltrated through the model's answer**Q12 B · obj 3.3****The tell:** ₹38 ... 2% ... ₹1.20 ... 98% ... ceiling ₹6**Why B:** $\text{Blended} = 0.02 \times 38 + 0.98 \times 1.20 \approx ₹1.94$ — already far under ₹6; the downgrade spends accuracy to fix a constraint that is not binding. Compute before you cut. The expensive tier on 2% of volume barely moves the blend; the review is reacting to the per-case number, not the constraint.**A [CM]** nothing needs raising; the blend is ₹1.94**C [VG]** added the per-case prices instead of weighting them**D [TR]** spends the currency the stem did not free up**Q13 D · obj 3.3****The tell:** 4 s SLA; p95 is 3.85 s ... +350 ms ... did not improve**Why D:** Reject — $3.85 + 0.35$ breaches the SLA at p95, and the measurement shows no accuracy to buy. Two reasons, either sufficient: the fixed currency (latency) is spent, and the dial has no measured gain past 20.**A [CM]** the SLA is the contract, not the variable**B [TR]** 20 beat 5 and 10 in measurement; do not throw away accuracy that fits**C [TR]** spends the fixed currency for nothing measurable

Q14 B · obj 3.3**The tell:** one after another ... SLA is 2 s**Why B:** Run the three calls concurrently — wall clock becomes the slowest scout (~1.1 s), inside the SLA. The SLA buys concurrency, not model speed. Three independent lookups have no reason to queue.**A [CM]** the data is needed; the sequencing is the problem**C [VG]** no model is involved in an HTTP call's latency**D [TR]** trades freshness the dispatcher needs for latency the fan-out would fix for free**Q15 A · obj 3.3****The tell:** floor of $\geq 99\%$... 97.5% ... 99.3%**Why A:** C — the floor is binding and B cannot clear it; 1.3 s sits comfortably inside 5 s. Bind the constraint first ($\geq 99\%$), then check the other currency has room. B fails the floor; C's latency is slack.**B [SD]** a retry does not raise a model's ceiling; it doubles its latency**C [CM]** 1.5 points below a stated floor is a breach, not a rounding error**D [CM]** nothing needs removing; $1.3\text{ s} < 5\text{ s}$ **Q16 A · obj 3.3****The tell:** which justification**Why A:** 'The refund path errs at ~₹1,800 per mistake and is 9% of volume, so it moves to the large model; blended cost stays $\leq ₹6$ '. Constraint, dial, direction, number. Everything else is an adjective with a price tag.**B [SD]** comfort is not a currency**C [VG]** a wish; no constraint, no number**D [SD]** practice is not a constraint**Q17 B · obj 3.4****The tell:** p50 = 200 ms and p99 = 6.1 s ... 'p99 under 3 s'**Why B:** The dashboard hides the metric the SLA is written in — add the p99 panel; the promise is breached at the tail. SLAs are promises about tails. A green p50 says nothing about the one-in-a-hundred who waits six seconds.**A [CM]** the SLA is the contract**C [SF]** the median is where tail pain hides**D [OE]** per-request alerts on a 1% tail is a siren every 40 seconds**Q18 D · obj 3.4****The tell:** 'retrieved chunks for ticket in 310ms'**Why D:** Convert logging to structured JSON with stage, ms and trace fields — a dashboard cannot aggregate prose. Metrics are computed FROM the register. Free-text lines cannot feed them; the fix is in the emitting code.**A [SF]** detects Tuesday's breach on Friday**B [SD]** regex over prose is a second, fragile parser for a problem the emitter should not create**C [SD]** dashboard theater — panels over data that does not exist**Q19 A · obj 3.4****The tell:** background worker that starts with a fresh context**Why A:** The trace id is not propagated across the worker boundary — pass it into the background call so its span joins the ticket's journey. A trace is only as complete as the hops that carry the id. The invisible span is the one that crossed a boundary without it.**B [VG]** retention keeps lines; it does not stamp them**C [TR]** throws away concurrency (3.3) to dodge a plumbing fix**D [SF]** 'not in the trace' is not 'not slow'**Q20 D · obj 3.4****The tell:** metric label set**Why D:** {stage, carrier, route} — a handful of categories, a handful of series. Categories in metrics; ids in logs and traces. Low cardinality is what keeps a metric cheap.**A [OE]** hundreds of series and a privacy problem; that question is a log query**B [VG]** one series per request: the cardinality trap by definition**C [OE]** 40,000 series a day; the metrics store melts

Q21 C · obj 3.4**The tell:** alert rule ... dispatcher-facing**Why C:** 'p95 total latency > 3.5 s sustained for 5 minutes' — a customer-felt symptom, few and dear. Alert on symptoms customers feel, sustained, at the tail. Everything else is a dashboard panel.**A [OE]** denials and handled retries are logged on purpose; this cries hourly**B [SD]** an internal wobble; a panel, not a page**D [OE]** pages on the 5% tail all day; the channel becomes noise**Q22 D · obj 3.4****The tell:** tokens and cost_pc ... stamped with route and stage**Why D:** Aggregate cost_pc by route and stage from the register — the growth localises to one route (the newly widened belt on the agent tier). Cost observability is the same plumbing as latency observability: per-stage fields, aggregated. Part 2's belt tax is visible only because Part 7 logged it.**A [SD]** the fields already exist; aggregate them**B [TR]** spends accuracy to avoid reading a log**C [VG]** the invoice knows totals, not your routes**Q23 C · obj 3.5****The tell:** no headings, dense legal prose**Why C:** Fixed-size with overlap and a situating sentence per chunk — with no structure to cut on, fixed is the argued-for default. Heading-aware needs headings. Amorphous prose walks the picker down to fixed-size + overlap, with contextual retrieval labelling every box.**A [CM]** clauses span sentences; the procedure is shredded**B [TR]** 3,000-token boxes for 40-token clause questions; precision dies**D [CM]** the stem says no headings — scanned prose has nothing to cut on**Q24 D · obj 3.5****The tell:** 'e-way bill' misses ... '□-□□ □□□'**Why D:** Write an English situating sentence into every chunk at ingest so BOTH indexes see the term, and let the semantic channel bridge the script. Contextual retrieval is how a mixed-script corpus becomes retrievable: the label inside the box carries the vocabulary the register needs.**A [TR]** throws away the exact-code half for a labelling problem**B [CM]** half the corpus is English; you have moved the miss, not fixed it**C [SF]** ships the vocabulary gap to the model and hopes**Q25 B · obj 3.5****The tell:** 60k-token ... stable ... nearly every call**Why B:** B — under the ~200k threshold, stable, always used: cache reads at ~90% off make it ~6k token-equivalents with no retrieval hop and no under-fetch risk. The pre-question of every RAG stem: is the corpus big enough to need RAG? 60k is not. Small + stable + always-used → cache.**A [TR]** reads only the token column; ignores the round-trip, the 180 ms and the risk of missing a clause**C [OE]** building the machinery before the corpus asks for it**D [VG]** caching is reliable above the tier minimum; 60k clears every minimum**Q26 C · obj 3.5****The tell:** retrieved at rank 7 ... only the top 5**Why C:** The fusion/depth stage, not shelving: pass top-20 (measured best) and add reranking — the right box is found, it is just not reaching the desk. Diagnose by where the paragraph fell out: 'right document, wrong rank' names fusion/rerank, not chunking.**A [TR]** shelving already works; the miss is downstream**B [TR]** a better foreman finds the same box at the same rank; the desk still takes five**D [SD]** persuasion downstream of a depth setting**Q27 B · obj 3.5****The tell:** 400 tokens on 500-token chunks ... near-duplicates**Why B:** Overlap is a dial and it is turned too far — ~80% duplication; set it to seam-insurance size (a few dozen tokens) and re-index. Overlap buys whole seam-straddling sentences and bills duplicate tokens. At 80% you pay twice and the shortlist fills with echoes.**A [SF]** more echoes, not more sections**C [TR]** a sentence across a seam then exists whole in no box**D [VG]** nothing indexed twice is free, and the shortlist proves it

Q28 B · obj 3.5**The tell:** repealed two weeks ago, with a confident citation**Why B:** Write a version label into every chunk's text at ingest and drop superseded chunks on re-index — staleness must be visible and removable. Indexes are caches of the corpus; a cache needs an invalidation story, and the citation must show its age.**A [SF]** the repealed procedure was cited fluently; the reader could not tell**C [OE]** per-edit pipelines for weekly-cadence documents**D [SD]** the model cannot see dates that were never written into the chunk**Q29 B · obj 3.5****The tell:** runs on every query**Why B:** Situating runs at INGEST — ~3,600 model calls per weekly rebuild, cacheable — not per query; the per-query cost is unchanged. The cost lands once, when the box is labelled, not each time it is fetched. 3,600 a week against 40,000 a day is the arithmetic that settles it.**A [VG]** the cut is real; the mechanism is wrong**C [VG]** tier changes the ingest price, not where it lands**D [VG]** nothing situates at query time**Q30 A · obj 3.6****The tell:** one row in a 12,000-row tariff table**Why A:** A structured lookup (or exact keyword match) against the table — an exact-answer question wants THE row, not similar rows. Rows with keys + an exact-entity question is the matrix's clearest cell. Similarity would return the batteries' neighbours.**B [SF]** a fluent wrong code on a customs form**C [OE]** converts certainty into probability; 'HSN 8506' embeds near 'HSN 8507'**D [OE]** excellent machinery in the wrong cell**Q31 D · obj 3.6****The tell:** trend ... per carrier this month**Why D:** RAG fetched chunks that MENTION holds and the model composed plausible numbers — 'trend', 'per', 'this month' is aggregate language: SQL over the TMS rows. RAG cannot count. Trend-and-per language is the router's aggregate tell, and a fluent wrong trend is the worst failure class.**A [SF]** citations prove the chunks exist, not that anyone counted**B [VG]** no k counts rows**C [OE]** the fashionable hammer; the TMS already answers exactly**Q32 A · obj 3.6****The tell:** keep getting stuck, and what should I do**Why A:** Decompose: an aggregate over TMS rows for 'keep getting stuck' (how often, since when) plus hybrid RAG for 'what should I do' — then answer from both. One question, two shapes, two machines. Feeding the whole thing to one route guarantees one half is wrong.**B [CM]** a lane is not a keyed row; the frequency half is a count**C [SF]** the 'keep getting' half becomes a fluent guess**D [OE]** a known decomposition at scale is workflow territory (D1)**Q33 A · obj 3.6****The tell:** on 14 Aug about KF-2481**Why A:** C — an id and a date are exact strings; the register's home game, and the metadata inside the chunk makes the filter possible. Query shape tunes the fusion: exact codes → keyword-lean. Situated metadata is what makes free-text corpora filterable.**B [SF]** summaries drop the exact sentence the dispute needs**C [TR]** the fuzzy part is small; the id and date decide the answer**D [CM]** raw chunks strand the id and date outside both indexes**Q34 B · obj 3.6****The tell:** no change ... cut failures 67%**Why B:** Keep reranking ON for prose paths and OFF for exact-code paths — dials are per path, and the measurement says where it pays. Reranking re-sorts a fuzzy shortlist; an exact match has nothing to re-sort. Spend the 200 ms where the 67% lives.**A [TR]** 200 ms bought nothing on one path**C [TR]** throws away a 67% failure cut where it fits**D [OE]** a bigger foreman for a path that needs the register

Q35 B · obj 3.6**The tell:** how many times ... KF-2481**Why B:** An aggregate query filtered by the id — SQL over the TMS rows WHERE consignment = KF-2481; count/trend language wins over the id pattern. The id picks the rows; the count is still a count. A lookup returns one row; the question wants a number over many.

- A [SF]** text about delays is not this consignment's delay count
C [VG] the far side negotiates slots; it does not keep your history
D [CM] a lookup returns the current row, not a count over history

Q36 C · obj 3.7**The tell:** six internal teams ... change monthly**Why C:** One MCP server: six clients dock once and discover the changed schema via tools/list — additions and changes propagate without client edits. Count clients and churn: both high. The container pays off at the pairing count.

- A [OE]** six welds, re-welded monthly
B [CM] six copies to update monthly re-creates the propagation problem
D [VG] nothing negotiates; these are function calls

Q37 D · obj 3.7**The tell:** changes every fortnight ... forty dispatchers keep pasting**Why D:** A prompt — the procedure lives once on the server as a versioned template the client fetches, instead of forty stale copies. Tools act, resources inform, prompts template. A reusable procedure is the prompt primitive's exact job.

- A [VG]** close, but a resource is data to read into context; a prompt is an interaction template the client applies
B [VG] elicitation asks the user for input; it does not distribute procedures
C [CM] spends the action primitive on a template need

Q38 B · obj 3.7**The tell:** control tower AND KAVI ... a third client is planned**Why B:** B — two clients today and a third planned is the pairing count that pays for the connector; the weld hides behind the standard once. The 3.7 sentence is about clients and churn, not about the vendor's API style. One endpoint, many callers → wrap it.

- A [TR]** pays three welds to defer one wrap
C [CM] the endpoint count is irrelevant; the client count decides
D [OE] standard for standard's sake on the single-client ones

Q39 D · obj 3.7**The tell:** confirm a pickup slot**Why D:** Elicitation — the client primitive that lets the server ask the user a question mid-task. Spec 2026-07-28: elicitation is current; sampling and logging are deprecated.

- A [VG]** logging is deprecated; log to stderr/OTel
B [VG] deprecated, and it asks the model, not the dispatcher
C [CM] the server has the question; the primitive exists so it can ask

Q40 D · obj 3.7**The tell:** authority limit ... turn budget**Why D:** Consignment, constraints, the authority limit ('may accept up to ₹900 surcharge'), a turn budget, and 'replies are proposals to verify'. Brief + budget + verification. The far side has judgement; give it goals and limits, not a function signature.

- A [CM]** no constraints, no authority, no budget — an open cheque
B [VG] a negotiation crammed into a lookup's shape
C [OE] the monolith crosses the company boundary

Q41 C · obj 3.8**The tell:** 10 of the 12 are used on almost every request**Why C:** Reject — nearly every call would pay fetch round-trips for 10 tools; keep the full schemas in the (cached) prefix: small, stable, always-used. The catalogue pays off when most of the belt is idle per request. When most of it is used every time, the round-trips are pure loss.

- A [CM]** trimming (3.1) is not discovery (3.8); the rare tools may be load-bearing
B [TR] leaner on tokens, heavier on turns, for tools that are always fetched anyway
D [VG] inverted: catalogue the rarely used, keep the always used

Q42 C · obj 3.8**The tell:** 150k-token ... only ~30% of calls**Why C:** C — 'always-used' fails: 70% of calls would carry 150k cached tokens for nothing; retrieval sends ~3.6k only when needed. The exception needs all three words: small, stable, AND always-used. Two out of three is the default, not the flip.**A [TR]** 15k token-equivalents × 28,000 idle calls a day is not small**B [SF]** summaries drop the clause the 30% needed**D [CM]** misses the third property; 70% of calls pay for a pack they never read**Q43 B · obj 3.8****The tell:** lost the get shipment status results**Why B:** Name that tool in `exclude_tools` so its results survive clearing, and let the memory tool save essentials before the trigger. `clear_tool_uses_20250919` has `exclude_tools` for exactly this; `memory_20250818` keeps essentials in files the trim cannot touch.**A [CM]** workable but ignores the platform mechanism the stem names**C [CM]** delays the loss; does not prevent it**D [TR]** reinstates the 150k+ bloat to protect one tool's results**Q44 C · obj 3.8****The tell:** 2,000-token ... Haiku-class ... zero hits**Why C:** 2,000 tokens is below Haiku 4.5's 4,096-token minimum cacheable prefix — nothing that small is cached on that tier. Caching minimums are per tier (1,024–4,096). Below the floor, the `cache_control` marker is a no-op.**A [VG]** placement is not the issue; size is**B [VG]** they do, above 4,096 tokens**D [VG]** it was never written**Q45 D · obj 3.8****The tell:** 30k-token stable rate card AND ... corpus 1.4M**Why D:** B — the rate card is small, stable and used by every case (cache it); the SOP corpus is large and case-specific (retrieve it) — each half gets its own rule. Two corpora, two shapes, two answers in one design. Law 5 and its exception, side by side.**A [CM]** 1.4M tokens fits no window**B [CM]** the cases need the card; that is the task**C [TR]** pays 400 round-trips for a card every case reads

Answer letters at a glance

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	A	10	C	19	A	28	B	37	D
2	A	11	C	20	D	29	B	38	B
3	A	12	B	21	C	30	A	39	D
4	A	13	D	22	D	31	D	40	D
5	D	14	B	23	C	32	A	41	C
6	C	15	A	24	D	33	A	42	C
7	C	16	A	25	B	34	B	43	B
8	A	17	B	26	C	35	B	44	C
9	A	18	D	27	B	36	C	45	D